Engine Model(s): PT6A

70-00-00

STANDARD PRACTICES - INSPECTION

1. General

A close and complete inspection is important to prolong engine life and give maximum performance. Check for loose or missing parts and inspect any engine part or component that has been worn or damaged. Damage to engine parts may result from improper clearance, lack of lubrication, undesired movement of parts which are bolted or pressed together, overloaded, uneven load distribution, heat, shock, extension of minor damage such as scratches, tool marks, grinding cracks, nicks, etc. Damage to engine parts may also result from presence of foreign matter such as grit, chips, moisture, chemicals, etc., or from incorrect techniques during removal and installation.

All inspection procedures should be done in a well lighted, clean and dust-free area. Benches should be clean to keep previously cleaned parts free of dirt and dust. All parts shall be suitably tagged to indicate necessary repair or replacement. Although most parts will require only a visual inspection, a certain number will require use of gages and other measuring equipment. Some damage may only be detected by magnetic particle or fluorescent penetrant inspection methods. Methods of inspection for specific parts and components are detailed in relevant sections of this manual.

2. Inspection

A. General

(1) Examine parts for alignment, distortion, foreign matter, looseness, out-of-roundness, sharp edges, scratches, taper, warping and wear. Additionally, check the following:

(a) Holes in cases, manifolds, pipes and tubes for obstructions.

(b) Gear teeth and splines for contact patterns.

(c) Magnesium parts for corrosion.

(d) Mounting pads, parting and seating surfaces, for smoothness and flatness. Use pencil carbon paper whenever a smear-type indication of surface smoothness is required.

(e) Plugs for tightness.

(f) Studs, dowels, and similar protruding parts for alignment and projection length.

(g) Protective surface coatings for completeness.

(h) Threads for condition.

B. Inspection Terms

(1) For definitions of the inspection terms used throughout this manual, refer to Table 601.

Table 601 Inspection Terms

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Term	Definition
Abrasion	Wearing away of small amounts of metal as a result of friction between parts.
Blister	The raised portion of surfaces caused by separation of layers of material.
Buckling	Large deformation from original shape. Usually caused by pressure or impact of a foreign object, unusual structural stresses, excessive heat, or any combination of these conditions.
Burning	Loss of material from excessive heat.
Burr	A sharp projection or rough edge.
Chafing	Wear caused by friction.
Corrosion	A surface chemical action resulting in surface discoloration, a layer of oxide, or, in advanced stages, the removal of surface material.
Crack	Pressure break in material
Dent	A smooth, round-bottomed depression.
Distortion	Change from the original shape.
Flaking	Loose particles of surface material.
Frosting	An initial stage of scoring caused by irregularities or high points of metal welding together with minute particles of metal transferring to the mating surface, giving a frosted appearance.
Galling	Fretting, and wearing away by friction.
Gouging	Removal of surface material, typified by rough and deep depressions.
Grooving	Smooth, rounded indentations caused by concentrated wear.
Inclusion	Foreign material enclosed in metal.
Metallization	Coating by molten metal particles sprayed through the engine.
Nick	Sharp bottomed depression with rough outer edges.
Peening	Flattening or displacing of material by repeated blows. A surface may be peened by continuous impact of foreign objects, or loose parts.
Pitting	Surface condition characterized by minute holes or cavities which occur on overstressed areas. The pits may occur in such profusion as to resemble spalling.
Scoring	Form of wear characterized by a scratched, scuffed, or dragged appearance, with markings in the direction of sliding.

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Term	Definition
Scratches	Narrow shallow marks or lines resulting from the movement of a metallic particle or pointed object across a surface.
Scuff	Dull or moderate wear of a surface resulting from slight rubbing.
Seizure	Welding or binding of two adjacent surfaces, preventing further movement.
Spalling	Surface or sub-surface damage characterized by chips of parent metal that spall or flake out of material leaving cavities of varying sizes and depths.
Stress-failure	Metal failure due to compression forces, tension, shear, torsion, or shock.
Tear	Parent metal torn by excessive vibration or other stresses.
Unbalance	Condition created in a rotating body by an unequal distribution of weight about its axis. Will result in vibration.
Wear	Condition resulting from a relatively slow removal of parent material. Frequently not visible to the unaided eye.

C. Inspection Gages

NOTE: When an inspection procedure requires a very accurate measurement, a micrometer, vernier caliper or a dial indicator must be used.

(1) If a micrometer or vernier is to be used, check gage for accuracy before making measurement. Be sure that contacting surfaces are clean and contacting faces of part to be measured are free of dirt and burrs. When using depth gages, ensure anvil is held tight and square against part to be measured.

(2) If a dial indicator is used, ensure indicator base is anchored firmly and that any swivel connections are tightened securely.

(3) When taking measurements with feeler gages, ensure final size of feeler is a reasonably snug fit.

D. Magnetic Particle Inspection

(1) The magnetic particle inspection is a nondestructive test procedure which detects cracks, voids, pits, subsurface holes and other imperfections. This method is applicable to ferromagnetic steel and has no effect on nonferrous materials.

E. Fluorescent Penetrant Inspection

(1) Fluorescent penetrant inspection is a nondestructive means of inspecting nonferrous or nonmagnetic materials for cracks, cold shuts, porosity and other defects having surface openings. This method may be used on ferromagnetic parts of complex structure which could give false indications when checked by the magnetic particle inspection methods.

F. Inspection of Fuel, Oil and Air Tubes

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NOTE: The following inspection criteria are a standard instruction. Specific instructions for tubing inspection, where applicable, will be described in the appropriate chapter within the manual.

(1) Circumferential scratches in tubes are acceptable, provided they do not exceed a 45 degree segment on circumference. Any number of scratches is permissible; these may be blended out, provided depth of blend does not exceed 0.005 inch.

(2) Longitudinal scratches in tubes must not exceed one-half inch in length. Any number of scratches is permissible.

(3) Nicks and chafing marks in tubes are acceptable and may be blended provided they do not exceed 0.005 inch in depth. Each blended area must not exceed one-half square inch.

(4) Dents in tubes are of limited acceptability, provided there are no restrictions to flow caused by sharp edges or corners. Dents are unacceptable within one inch of ferrule scarf-welds, or in a blended area.

(5) Minor isolated pitting is acceptable if not greater than 0.003 inch in depth. Clusters of pits should be blended out to a maximum depth of 0.005 inch.

(6) Rust and stains on tubes are acceptable if they can be removed by light polishing with very mild abrasive.

(7) Examine tubes for cracks, especially at brazed joints.

(8) Examine coupling and threaded fittings for cracks or damaged threads.

(9) Check tube bores for obstructions.

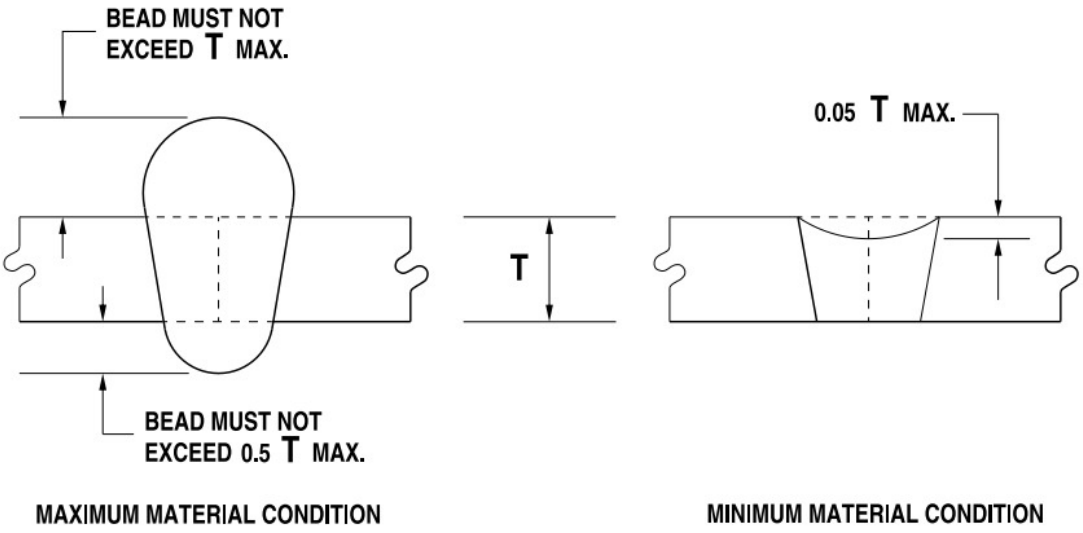
G. Inspection of Welded Repairs

(1) Inspect weld repair for quality, uniformity, undercutting, cracking and flux removal. Welds must be blended into adjacent metal in gradual, smooth curves. Welds must be sound, clean, free from foreign material, and from internal and external defects which would adversely affect strength of weld.

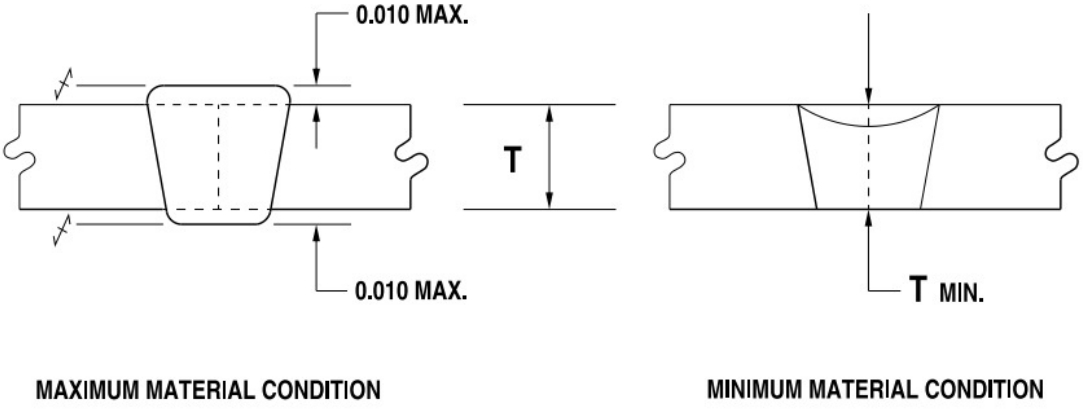
(2) Nominal welds (those not requiring finishing) must exhibit bead conditions as shown in Figure 601, detail A. Finished welds must exhibit bead conditions as shown in detail B.

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Figure 601 Weld Repair - Inspection



A – NOMINAL WELD – NO FINISHING REQUIRED



B – FINISHED WELD – DIMENSIONS AFTER DRESSING

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H. Inspection of Rotating Components

(1) Life limited (critical) rotating components exhibiting cracks are unserviceable and shall not be repaired.

(2) Handle rotating components with extreme care to prevent damage or contamination of the material.

(3) Wear lint free gloves when touching rotating components.

(4) Make sure close attention is paid to areas such as counterweight holes and blade slots where imperfections are difficult to find.

(5) Use suitable clean, protective containers for storing and transporting components

(6) Inspect components in accordance with the criteria specified in the relevant Chapter/Section/Subject.

(7) Any observed conditions not defined in this manual are not acceptable for repair.

(8) Evaluation of the surface should take into consideration the need for repairs and the limits to which repairs may be made to determine the suitability for continued use of the component.

(9) Unrepairable components shall be quarantined to prevent further use.

(10) It is important to have suitable lighting. The minimum recommended illumination for close and difficult inspection is 100 foot-candles measured at the inspection table top.

(11) Unless otherwise specified, magnification shall not be used. A magnification of 3X to 4X may be used as an aid to evaluate and confirm an observed condition in detail.

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